Project Update Discussion

12 November 2024 21:02

200GB GPU → D.L models

slow

matrix multiplic^n

TM → 200GB GPU

Agenda → (1) (Guidance on Capstone Project)

(2) → 1

(2) Portfolio (Rest of the crowd)

(3) Integrating Retail (Analytics) → Resume

Projects

CRM analysis

Price optimz^n Case Study

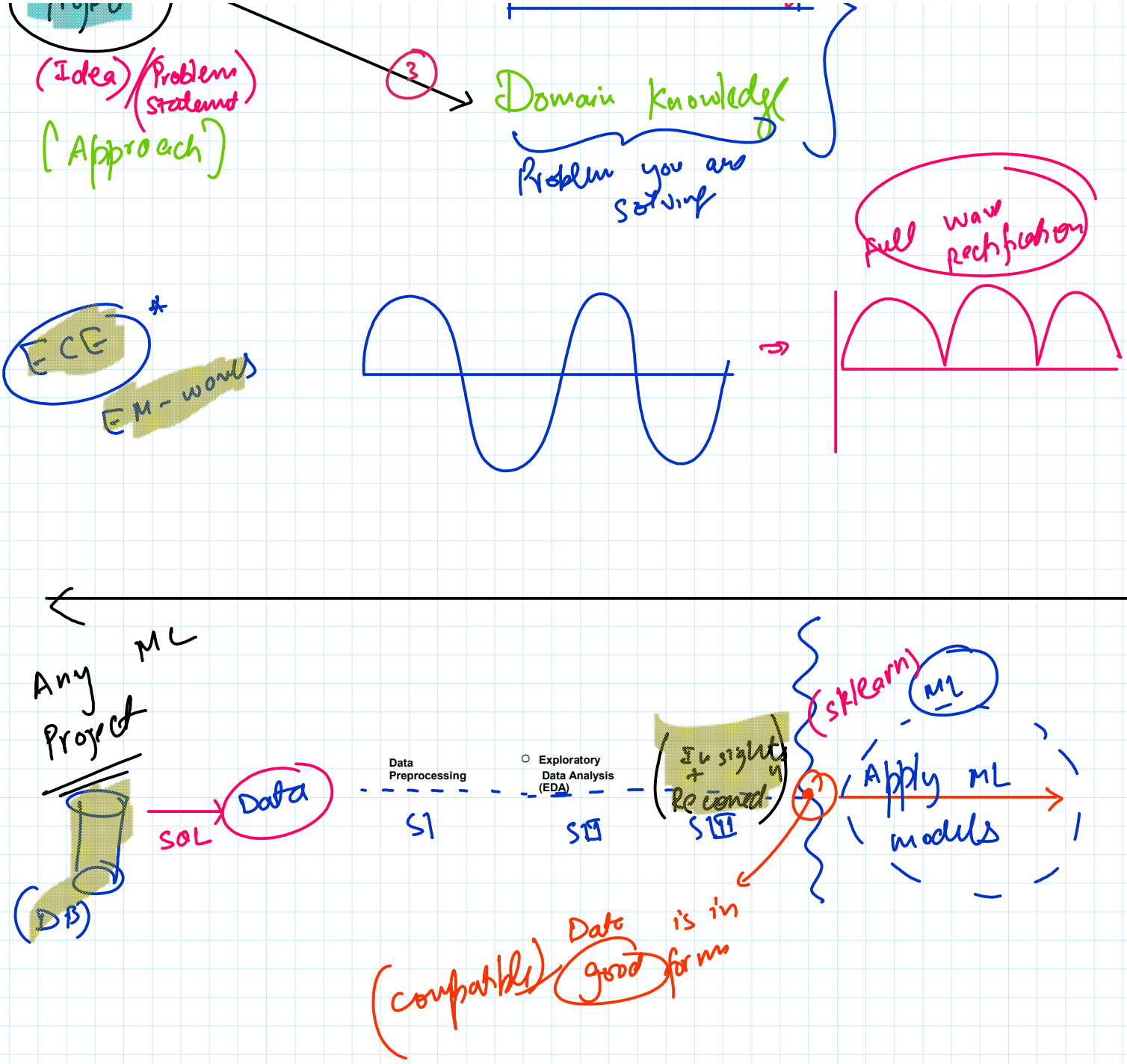
Project (Idea) (Problem)

(1) → Applied Knowledge (code) → (upload) (Github)

(2) → Core (D.S/DA) Theoretical Knowledge

(3)

Tested in Project



Guide (Project)

S1

- **Data Preprocessing**
  - Handling Missing Values: Start by identifying any missing values in the dataset, especially in key columns like CustomerID or InvoiceNo. Develop strategies to address missing data, such as imputing values or removing incomplete records.
  - Dealing with Duplicates: Check for duplicate rows in the dataset and remove any redundant entries to maintain data integrity.
  - Outlier Treatment: Handle outliers in the dataset, such as transactions with extremely high or low quantities or prices, which could distort your analysis. Pay special attention to negative quantities indicating returns and zero-priced items.
  - Data Type Correction: Ensure that each column has the correct data type for analysis. For

indicating returns and zero-priced items.

- Data Type Correction: Ensure that each column has the correct data type for analysis. For example, InvoiceDate should be in date-time format for time-series analysis.

SII Hind E. D.

- **Exploratory Data Analysis (EDA)**

  - **Exploratory Data Analysis (EDA)**
    - Descriptive Analysis: Begin by summarizing the key characteristics of the dataset, such as the number of unique customers, products, and countries. Visualize the distribution of transactions over time and look for patterns, seasonality, or spikes in customer activity.
    - Analyze Product Popularity: Examine which products are frequently purchased and identify any trends in customer preferences.
    - Customer Behavior Analysis: Investigate customer purchasing behavior by analyzing metrics such as average order value, order frequency, and time between purchases.
    - Return Transactions: Conduct an analysis of return transactions by filtering negative quantities. Understand which products have the highest return rates and explore the reasons behind these returns.

- Any feature dependent EDA
- Histogram / Box Plot / scatter Plot

SIII Feature Engineering → intelligence to add new columns
(helping)

→ (models lif easy by already including this)

Pattern

x_1	x_2	x_3	weekend or weekday	Fraud → Yes / No

  - **Feature Engineering**
    - RFM (Recency, Frequency, Monetary) Analysis: Create customer-centric features by calculating Recency (days since the last purchase), Frequency (number of purchases), and Monetary value (total amount spent by a customer). Use these features to segment customers into groups, such as high-value or at-risk customers.
    - Transaction Time Features: Generate new features that provide insights into customer behavior, such as average days between purchases, preferred days of the week for shopping, and peak shopping hours.
    - Cohort Analysis: Perform cohort analysis to understand how different groups of customers behave over time. Group customers based on the month of their first purchase and track their subsequent purchasing behavior.

CRM Project

  - **CRM-Specific Metrics and Analysis**
    - Customer Segmentation: Use the RFM model to segment customers into categories like "Loyal Customers," "New Customers," and "Churned Customers." Each segment should be analyzed separately to understand their behavior and devise specific strategies for engagement.
    - Lifetime Value (CLV) Calculation: Estimate Customer Lifetime Value (CLV) for individual customers

CRM Proj

Real world applic" of CRM

Customer Segmentation: Use the RFM model to segment customers into categories like "Loyal Customers," "New Customers," and "Churned Customers." Each segment should be analyzed separately to understand their behavior and devise specific strategies for engagement.

- Lifetime Value (CLV) Calculation: Estimate Customer Lifetime Value (CLV) for individual customers or segments. This metric helps identify high-value customers who are crucial to business profitability.
- Churn Prediction (Qualitative Analysis): Without using machine learning, explore churn indicators such as the recency and frequency of purchases. Identify patterns that suggest customer disengagement and propose strategies for re-engagement.

- **Insights and Recommendations**
  - Identifying Key Drivers: Use your analysis to identify key drivers of customer behavior, such as product preferences, purchase timing, or return trends. These insights can inform product recommendations, marketing strategies, and customer retention efforts.
  - Strategic Recommendations: Based on your findings, offer actionable recommendations to the business. For example, suggest personalized marketing campaigns for high-frequency customers or strategies to reduce return rates for specific products.
  - Performance Improvement: Propose how the business can use the insights gained to enhance its overall CRM strategy, including customer retention, satisfaction, and sales growth.

(P.O)

Real world Applica"

- **Real-World Application**
  - **Maximizing Profitability:** By leveraging price optimization techniques, businesses can ensure they are charging prices that lead to higher profits without alienating their customer base. The right pricing model accounts for costs, demand, and competition, allowing for strategic price adjustments.
  - **Competitive Advantage:** In a highly competitive retail landscape, offering the best price for products without compromising on profit margins is key to attracting customers. Price optimization allows businesses to monitor competitors and adjust prices dynamically to stay ahead.
  - **Improving Inventory Management:** Price optimization helps balance supply and demand, ensuring that products are sold at the right price to avoid overstock or understock situations. This is especially crucial for managing inventory costs and ensuring that popular items remain available.
  - **Adapting to Seasonal Trends:** In the fashion industry, where trends shift rapidly, pricing strategies must adapt to seasonal variations and product lifecycles. Using price optimization models, companies can adjust their prices to reflect the value of trendy items during high-demand periods or markdowns at the end of a season.
  - **Enhancing Customer Satisfaction:** Finding the right price point ensures customers feel they are getting value for their money, which can increase customer loyalty, repeat purchases, and overall customer lifetime value.

Submission Requirements

(1) ppt (5 – 8 slides)

→ Conclusion

Intro → (4 - 6) slides → Conclusion ($)

(2) Record a Loom video (5 minutes)

Why a Presentation?

- Creating a presentation is an essential skill for data professionals. It helps you distill complex analyses into clear, actionable insights that stakeholders can easily understand.
- The presentation forces you to focus on the most critical aspects of your work, enhancing your ability to communicate effectively.

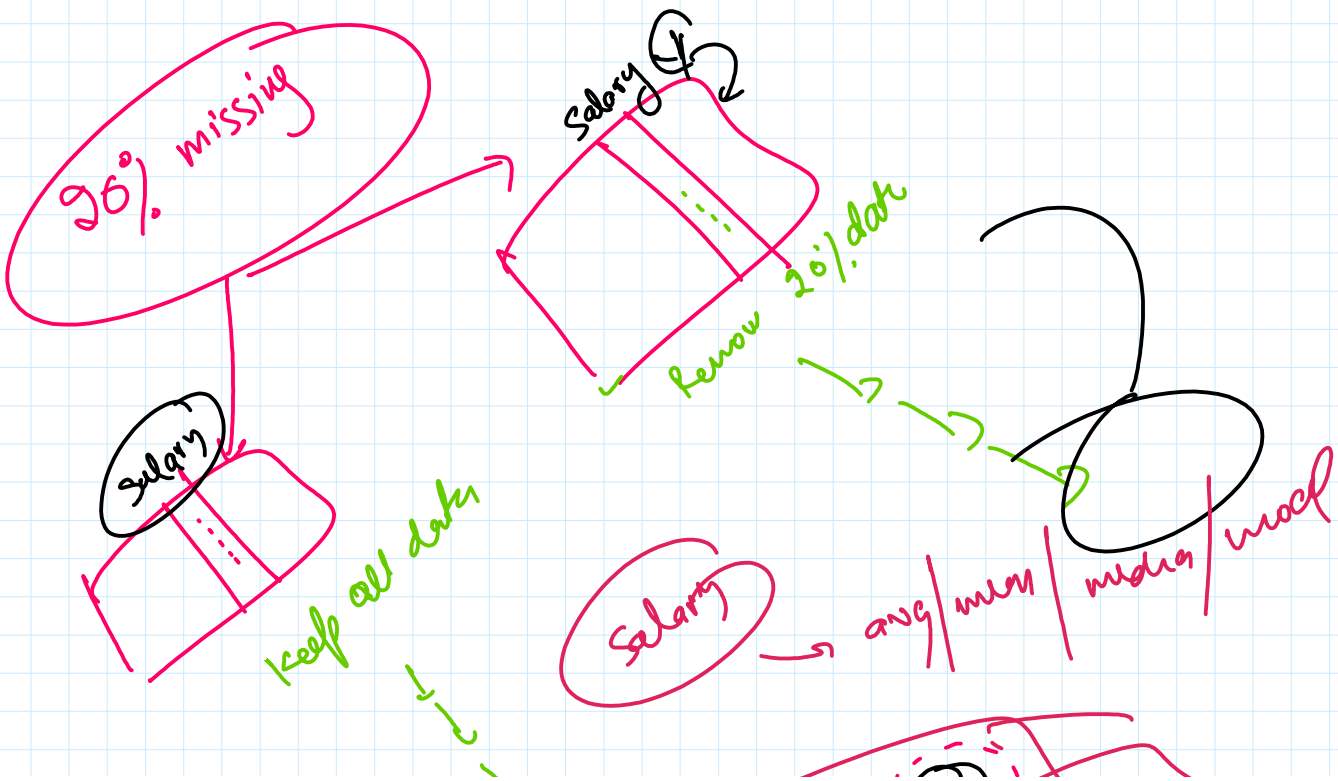
What Should Be in the Presentation?

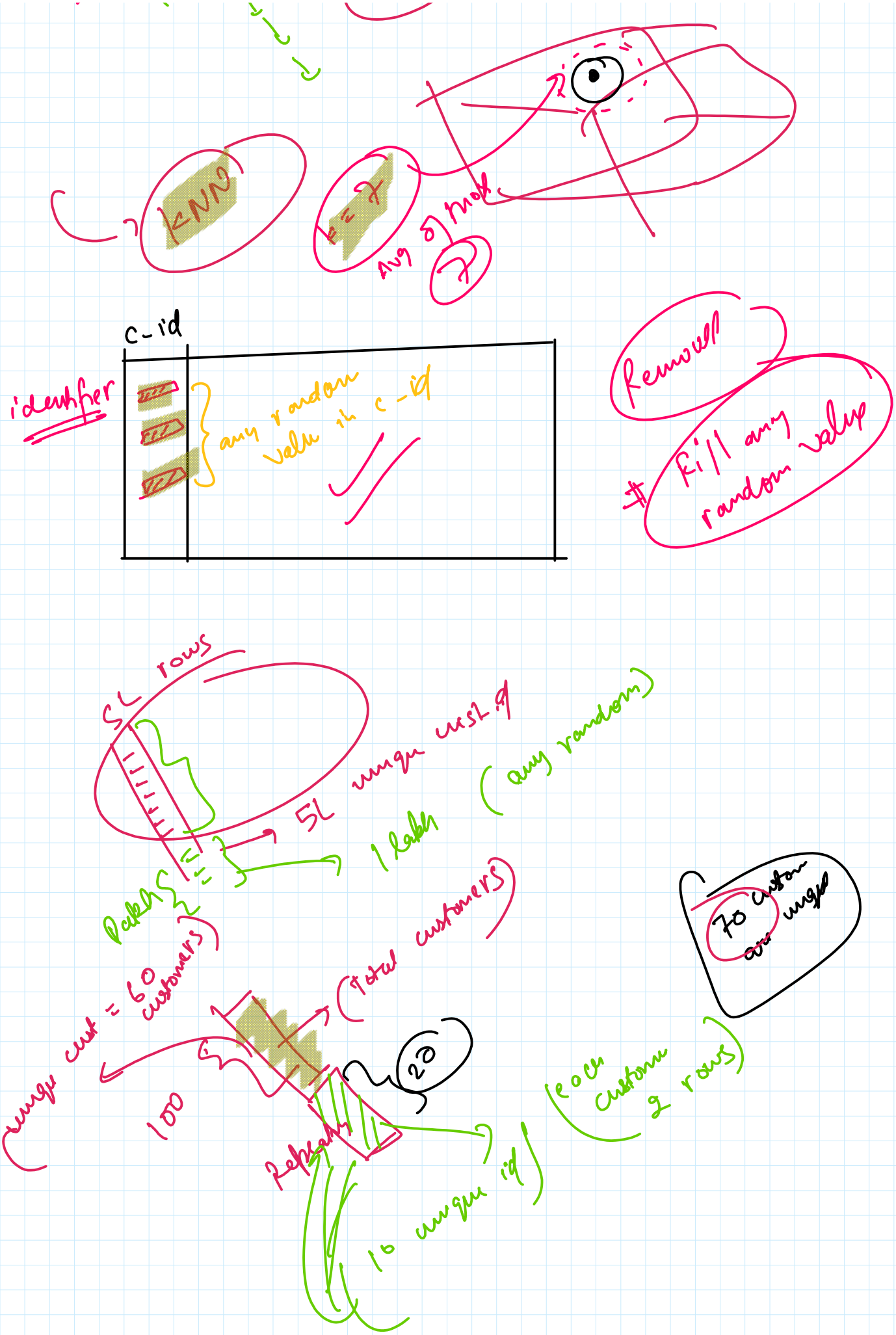
- **Introduction Slide:** Set the stage by introducing the project. This helps your audience understand the context and relevance of your analysis.
- **Summary of Important Observations:** Use visualizations to summarize key findings. This is your opportunity to highlight the most significant insights from your analysis, making it easier for stakeholders to grasp the data's story.
- **Recommendations Slide:** Based on your analysis, provide actionable recommendations. This is where you demonstrate your ability to not just analyze data but also to translate that analysis into strategic business decisions.

Scaler Study (E-Y)

Data Science Portfolio

(7 min break) → 21:49 *




KNN
k = 7
Avg of most
7
c_id
identifier
any random value in c_id
Remove
Fill any random value
50 rows
50 unique cust id
(any random)
1 label
Repeats
(total customers)
unique cust = 60 customers
70 custom are unique
100
20
Repeats
(each customer 2 rows)
10 unique id

Personal website) → (how a simple website)

- Domain name
- Hosting Service
- (Code) (HTML CSS JS)
  - ↳ Bootstrap templates
- (Github Pages)

→

Data Science Portfolio → (Code that you write)

→ (Cloud)

datascienceportfol.io

↳ simple easy mode sol^n (free)

↳ upload Projects

↳ (Directly paste the URL in Resume)

↳ D. scientist at ZS (GGN) Delhi-NCR

datascienceportfol.io/nishantlokhande

www.datascienceportfolio.io/username

www.datascienceportfolio.

Resume Bullet Points (Resume)

Visit the dashboard sec of today's session (scribble + Resume Pointers)

1 project → 1 article (Resume)

→ (combine 2 projects → (write as one))

→ upload on github / ds pf.io etc. etc.

List → include (info condensed into one)

Summary

1. Projects
   - T
   - A
   - D

   2x 3x 4x 5x
2. Projects Update Case Study
3. ds9.io → current / Past
4. Resume Pointer → DOC

equipped with

All the Best

All the Best

(#) Concluded